

Maximum likelihood estimation

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Logistics and reminders

- + HW 1 due tonight
- + Reminder: bank of 5 extension days -- can use 1 or 2 days on a homework assignment or project piece
- + HW 2 released, due next Friday
- + Project part 1 released (choosing a dataset + EDA)
 - + Project will be released in several pieces
 - + Part 1 due in 2 weeks
- + Coming up:
 - + Logistic regression assumptions and diagnostics
 - + Inference (hypothesis tests, confidence intervals)

Warmup: maximum likelihood estimation

Work on the activity (handout), then we will discuss as a course

Warmup

Independent observations $Y_1, \dots, Y_n$ with
 $P(Y_i|\pi) = (1 - \pi)^{Y_i-1}\pi$.

Likelihood function:

$$L(\pi) = P(Y_1|\pi)P(Y_2|\pi) \cdots P(Y_n|\pi) = \prod_{i=1}^n P(Y_i|\pi)$$

Warmup

$$L(\pi) = (1 - \pi)^{\sum_{i=1}^n Y_i - n} \pi^n$$

MLE for logistic regression

Suppose we observe independent samples $(X_1, Y_1), \dots, (X_n, Y_n)$ with

$$Y_i \sim \text{Bernoulli}(\pi_i) \quad \log\left(\frac{\pi_i}{1 - \pi_i}\right) = \beta_0 + \beta_1 X_i$$

Likelihood:

Class activity, Part 2

Maximizing likelihood for a logistic regression model in R (linked on class website):

https://sta214-f26.github.io/class_activities/ca_06_2.html

(You do not need to submit this part of the activity)